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United States Patent Application Publication

20250277379

Kind Code

A1

Publication Date

September 04, 2025

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Walkboard and End Cap

Abstract

A walkboard for a scaffold comprises one or more planks forming a platform for standing and one or more end caps pressed into each end of the platform. The end caps include integral hooks for engagement with horizontal members of the scaffold. The planks and end caps are specially designed to enable compact stacking for storage and shipment.

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Family ID: 96881004

Appl. No.: 18/591839

Filed: February 29, 2024

Publication Classification

Int. Cl.: E04G5/08 (20060101); E04G7/34 (20060101)

U.S. Cl.:

CPC E04G5/08 (20130101); E04G7/34 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to the field of scaffolding and, more particularly, to the design and construction of walkboards for use with scaffolding.

BACKGROUND

[0002] Scaffolds are used in the construction industry to provide an elevated work surface for a

task such as painting, installing trim, drywall finishing, etc. A typical scaffold comprises two spaced-apart frames interconnected by cross-braces to form a self-supporting structure. Walkboards extend between the frames to provide an elevated surface on which a worker stands. The walkboards typically include hooks at each end that engage with horizontal frame members on the frames.

[0003] One common type of walkboard comprises two interlocking planks connected edge-to-edge to form a platform for standing. The planks are made of aluminum using an extrusion process. Each plank includes a top wall and a plurality of longitudinally extending ribs projecting down from the top wall. Hooks are bolted to each end of the platform for engagement with the frame of the scaffold to support the platform in an elevated position. For mounting the hooks, a flat cross-bar is attached to the underside of the platform. The bolts securing the hooks pass through aligned openings in the top wall of the planks and the cross bar. The head of the bolt extends above the top plane of the walkboard and the end of the bolt to which the nut is threaded extends below the bottom plane of the walkboard. This assembly method greatly increases the overall height of the final assembly and complicates stacking of the walkboards for storage and shipment.

SUMMARY

[0004] The present disclosure relates to an end cap for walkboard and method of attachment that greatly reduces the overall height of the assembled walkboard. The height reduction enables more compact packaging of the walkboard for storage and shipment, which in turn, reduces the overall cost of the walkboard to the end user.

[0005] In an exemplary embodiment, the walkboard comprises one or more planks made of aluminum using an extrusion process. In embodiments having two or more planks, the planks include interlocking features to enable the planks to be connected edge-to-edge to form a platform for standing. The platform includes a top wall with an upper surface and a plurality of longitudinally extending ribs extending downward from the bottom of the top wall. The ribs define a series of longitudinally extending channels between adjacent pairs of ribs that are open at the opposing ends of the platform.

[0006] An end cap is configured for attachment to the opposing ends of the platform. Each end cap includes a face plate configured to cover two or more of the open-ended channels in the one or more planks, a set of two or more plug bodies extending from a back surface of the face plate, and one or more hooks extending from a front surface of the face plate and configured to engage a horizontal frame member of the scaffold. The plug bodies are separated by a gap and configured to fit into adjacent channels in the platform. The gap between the plug bodies receives the rib separating the adjacent channels when the end cap is pressed onto the end of the platform.

[0007] In some embodiments, a first plug body includes an access bay extending inward from a bottom surface of the plug body. The access bay includes a first angled surface with a first mounting hole formed therein. The second plug body may include a second angled surface with a second mounting hole axially aligned with the first mounting hole. In this embodiment, a threaded bolt or other elongate fastener extends through the aligned mounting holes, passing through the rib separating the adjacent channels. A nut threads onto the end of the bolt to secure the end cap to the platform.

[0008] In some embodiments, the end cap includes at least two hooks extending from the front surface of the face plate. The hooks are spaced along the face plate such that the hooks on a first walkboard in a non-inverted orientation are offset from the hooks on a second walkboard when the second walkboard is inverted for stacking on the first walkboard. This arrangement of the hooks enables the walkboards to be vertically stacked without the need to stagger the walkboards.

[0009] In some embodiments, the end cap includes a notch along a lower edge of the face plate. A safety latch attaches to the bottom surface of the second plug body and extends through the notch in the face plate to prevent the walkboard from lifting off the horizontal member of the scaffold when the walkboard assembly is in use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0010] FIG. **1** is a perspective view of a scaffold and walkboard.
[0011] FIG. **2** is a perspective view of a walkboard.
[0012] FIG. **3A** is a cross-section of the walkboard with the end cap removed.
[0013] FIG. **3B** is a close-up cross-section of the walkboard with the end cap removed.
[0014] FIG. **4** is an exploded perspective view of a walkboard.
[0015] FIG. **5** is a top view of an end cap for the walkboard.
[0016] FIG. **6** is a front view of an end cap for the walkboard.
[0017] FIG. **7** is a rear view of an end cap for the walkboard.
[0018] FIG. **8** is a side view of an end cap for the walkboard.
[0019] FIG. **9** is a partial perspective view showing one end of the walkboard from below.
[0020] FIG. **10A** is a cross-section of the walkboard with the end cap installed.
[0021] FIG. **10B** is a close-up cross-section of the walkboard with the end cap installed.
[0022] FIG. **11** is a side view showing one end of the walkboard.

DETAILED DESCRIPTION

- [0023] Referring now to the drawings, FIG. **1** illustrates an exemplary scaffold **10** and walkboard **30**. The scaffold **10** comprises two spaced-apart frames **12** interconnected by cross-braces **24** to form a self-supporting structure. Each frame **12** includes two vertical posts **14** interconnected by horizontal members **16**. A connecting member **18** extends vertically between the horizontal members **16** approximately midway between the vertical posts **14**. Ladder rungs **20** extend between the connecting member **18** and one of the vertical posts **14** to form a ladder.
- [0024] The components of the frame **12** are made of metal tubing which is welded together to form a single piece. The vertical posts **14** include stud posts **22** with latches for engagement by the cross-braces **24** as well-known in the field of scaffolding. The latches retain the cross-braces **24** on the stud posts **22** when the cross braces **24** are engaged with the stud posts **22**.
- [0025] One or more walkboards **30** extend between the frames **12** of the scaffold **10** to provide an elevated platform on which a worker stands. The walkboards **30** include hooks **64** at each end that engage with the horizontal members **16** on the frames **12** to support the walkboards in an elevated position. During use, a safety latch **90** (FIG. **11**) can be installed to prevent the walkboards **30** from inadvertently lifting up off of the horizontal members **16** and becoming a tipping hazard.
- [0026] FIG. **2** illustrates an exemplary walkboard **30** in more detail. The walkboard **30** generally comprises one or more planks **34** connected edge-to-edge to form a platform **32**, and a pair of end caps **60** pressed onto opposing ends of the platform **32**. The end caps **60** include integrally formed hooks **64** that engage with the horizontal members **16** of the frames **12** as previously described.
- [0027] FIGS. **3A** and **3B** illustrate one embodiment of a platform **32** comprising two interconnected planks **34**, which can be formed via extrusion of a lightweight material, such as aluminum. For purposes of the following discussion, reference number **34** is used to refer to the planks generally, while reference numbers **34A** and **34B** are used to differentiate between the planks **34** when necessary.
- [0028] Each plank **34** includes a top wall **36** having an upper surface and a bottom surface and a plurality of longitudinally-extending ribs **38** projecting down from the top wall **36**. The ribs **38** extend longitudinally along the length of the plank **34**, i.e., parallel to the longitudinal axis of the plank **34**. In the illustrated embodiment, each plank **34** includes three ribs **38**: an outer rib **38A**, an interior rib **38B**, and an inner rib **38C**. The outer rib **38A** of each plank **34** forms an outer wall of the platform **32** and has a generally L-shaped cross section with a flange **40** extending inward from the lower end of the outer rib **38A**. The interior rib **38B** of each plank **34** has a T-shaped cross-section with a flange **42** at the lower end of the rib **38B** that extends outward from each side of the

interior rib **38B**. The inner ribs **38** form the inner walls of the planks **34** that abut when the planks **34** are assembled edge-to-edge.

[0029] The planks **34A** and **34B** are designed with interlocking features that enable the planks **34A** and **34B** to slide together longitudinally during assembly. These interlocking features are shown more clearly in FIG. 3B. The interlocking features provides two points of engagement: one at the lower end of the inner ribs **38C** and one adjacent the top surface **36**. The complementary features enable the planks **34A** and **34B** to be assembled by relative sliding motion after aligning the interlocking features on the two planks **34A**, **34B**.

[0030] A first set of interlocking features comprises a flange **44** that extends outwardly away from the lower end of the inner rib **38C** on one plank **34** (plank **34B** in this example) and a complementary channel **46** formed at the lower end of inner rib **38C** of the other plank **34** (plank **34A** in this example). The flange **46** is designed to slide longitudinally within the channel **46** when the planks **34A** and **34B** are being assembled while restricting relative vertical and relative side-to-side motion.

[0031] A second set of interlocking features comprises a tongue and groove connection **50** formed in the top surface **36** of the planks **34**. The tongue and groove connection **50** includes a tongue **52** on one plank **34** (plank **34A** in this example) that projects out horizontally beyond the inner rib **38C** and a complementary groove **54** formed in the other plank **34** (plank **34B** in this example) to receive the projecting tongue **52** of plank **34A**. The tongue and groove connection **50** is designed so that the tongue **52** slides longitudinally within the groove **54** when the planks **34A** and **34B** are being assembled while restricting relative side-to-side motion. The tongue **52** includes a channel behind the head of the tongue **52**. This channel receives the upper end of inner rib **38C** on plank **34B**.

[0032] To assemble the platform **32**, the planks **34A** and **34B** are placed end-to-end and offset so that interlocking features of the planks **34** align. The planks **34** will then slide together longitudinally. Afterwards, the end caps **60** are installed, which prevent relative longitudinal movement of the planks **34**. Any number of planks **34** can be assembled together to form a platform of any desired width. The illustrated embodiment includes two planks **34**, but other embodiment could include 3 or more planks **34**.

[0033] FIGS. 4-8 illustrate the end cap **60**. The end cap **60** comprises a face plate **62**, one or more hooks **64** extending from a front surface of the face plate **62**, and a plurality of plug bodies **70** extending from the back of the face plate **62**. The end cap **60** is designed to press onto the end of the platform **32** after the planks **34** are connected. As shown in FIG. 4, the plug bodies **70** are separated by gaps **72** and designed to fit into the open-ended channels in the platform **32**. The gaps **72** between the plug bodies **70** provide space for the ribs to slide into when the end cap **60** is pressed onto the end of the platform **32**.

[0034] FIGS. 4-8 illustrate one embodiment of an end cap **60** that extends the entire width of the walkboard **30** so only one end cap **60** needs to be installed at each end of the walkboard **30**. In this embodiment, the face plate **62** extends the full width to the platform **32** and covers all four of the open-end channels in the platform **32**. This embodiment includes two hooks **64** extending from the front of the face plate **62** and four plug bodies **70** extending from the back of the face plate **62**.

[0035] In other embodiments, two or more end caps **60** could be installed at each end of the walkboard **30**. As an example, one embodiment of the walkboard **30** could include two end caps **60** at each end of the walkboard **30** (four in total), with each end cap including one hook **64** and two plug bodies **70**. One of the end caps **60** could be designed to press onto plank **34A** and the other end cap **60** could be designed to press onto plank **34B**.

[0036] Returning to FIGS. 4-8, the unitary end cap **60** includes two hooks **64**. One hook **64** is disposed adjacent one side edge of the walkboard **30**. The other hook **64** is offset from the opposing side edge. This offset enables the assembled walkboards **30** to be stacked more compactly by inverting one of the walkboards **30**. In a mini-stack of two walkboards **30**, the first walkboard **30** is

placed in an upright position. It will be noted that the curvature of the hooks **64** extend above the plane of the top plane of the walkboard **30**, which is defined by the top surface of platform **32**. The second walkboard **30** is inverted and stacked on the first walkboard **30**. Because of the offset, the hooks **64** do not interfere when the walkboards **30** are stacked. A larger stack can then be made by stacking the mini-stacks on top of each other up to any desired height.

[0037] The plug bodies **70** are also designed so that the plug bodies **70** will not extend below the bottom plane of the walkboard **30**, which is defined by the lower end of the outer ribs **38A**. Each of the plug bodies **70** has a generally rectangular configuration with a top wall, **70A**, bottom wall **70B**, and side wall **70C** and **70D**. The overall outer shape provides a friction fit with the matching channels in the platform **32**. The plug bodies are hollow and open towards the back so that the interior of the plug bodies **70** is accessible from the back.

[0038] Conceptually, there are two pairs of plug bodies. Each pair comprises an outer plug body **70** adjacent one side edge of the platform **32** and the adjacent inner plug body **70**. The two outer plug bodies **70** adjacent the opposing sides of the walkboard **30** have an access bay **74** formed in the bottom wall **70B** with an angled surface **76** that is part of the side wall **70D**. The two inner plug bodies **70** include an angled interior surface **78** that is part of the side wall **70C**. For each pair of plug bodies **70**, axially aligned openings **80** and **82** are formed respectively in the angled surfaces **76** and **78** for a bolt that secures the end cap **60** to the platform **32**. During installation, a hole is drilled through the intervening rib **38** between the outer and inner plug body **70** in each pair to allow installation of the bolt **84**. A nut **86** is threaded onto the end of the end of each bolt **84** and tightened to secure the end cap **60** to the platform **32**. The installation of the end cap **60** prevents relative longitudinal movement of the planks **34**.

[0039] In order to prevent the walkboard **30** from lifting off the horizontal member **16** of the scaffold **10**, a safety latch **90** is attached to one of the plug bodies after the walkboard is set into place on the scaffold **10**. As shown in FIG. **11**, the safety latch **90** extends below the horizontal member **16** of the scaffold **10** preventing the walkboard **30** from lifting up. The safety latch **90** is secured to one of the inner plug bodies **70** by a bolt **92** and nut **94**. When secured in place, the safety latch **90** extends through a notch **66** in the face plate **62** so that the safety latch **90** stays above the bottom plane of the walkboard **30** and does not interfere with compact stacking.

[0040] In use, a platform **32** is assembled from two planks **34** and two end caps **60**. The planks **34** are placed end-to-end and offset so that interlocking features of the planks **34** align. The planks **34** will then slide together longitudinally. Afterwards, the end caps **60** are pressed into each end of the platform **32** and secured by bolts **84** and nuts **86** as previously described. During installation, a hole needs to be drilled through the ribs **38** separating the outer plug bodies **70** from the adjacent inner plug bodies **70**. Once assembled, the walkboard **30** can be placed on the scaffold **10** with the hooks **64** at each end engaged with a horizontal member **16** of the scaffold **10**. Finally, to prevent the walkboard **30** from lifting up off the horizontal member **16**, the safety latch **90** is installed at each end of the walkboard **30** to hold the walkboard **30** in place.

Claims

1. A walkboard for use with a scaffold, the walkboard comprising: one or more planks forming a platform for standing, wherein the platform includes a top wall and a plurality of longitudinally extending ribs projecting downward from the top wall, and wherein the ribs define a series of longitudinally extending channels between adjacent pairs of ribs that are open at the opposing ends of the platform; an end cap configured for attachment to each end of the platform, each end cap comprising: a face plate having a front surface and a back surface and configured to cover two or more of the open-ended channels in the platform forming the platform; a first plug body having a top surface, a bottom surface, and two sidewalls extending from the back surface of the face plate and configured to fit into a first channel in the platform, the first plug body including an access bay

extending inward from the bottom surface and having a first angled surface with a first mounting hole formed therein; a second plug body having a top surface, a bottom surface, and two sidewalls extending from the back surface of the face plate and configured to fit into a second channel in the platform adjacent to the first channel; a gap between the first and second plug bodies configured to receive the rib between the adjacent channels when the end cap is pressed onto the end of the platform; at least one hook extending from the front surface of the face plate and configured to engage a horizontal member of the scaffold; an elongate fastener for securing the end cap to the platform, the elongate fastener configured to extend through the first mounting hole in the first plug body and at least partially into a first rib of the plurality of ribs separating the first and second channels.

2. The walkboard of claim 1, wherein the second plug body includes a second angled surface with a second mounting hole, the second angled surface being disposed generally parallel to the first angled surface in the first plug body with the second mounting hole axially aligned with the first mounting hole.

3. The walkboard of claim 2, wherein the elongate fastener comprises a bolt configured to extend through the first and second mounting holes and a nut configured to thread onto one end of the bolt to fasten the end cap to the first rib separating the first and second channels.

4. The wallboard of claim 1, wherein the end cap includes at least two hooks extending from the front surface of the face plate.

5. The walkboard of claim 4, wherein the hooks are spaced along the face plate such that the hooks on a first walkboard in a non-inverted orientation are offset from the hooks on a second wallboard when the second walkboard is inverted for stacking on the first walkboard.

6. The walkboard of claim 1, further comprising: a notch in a lower edge of the face plate; a safety latch configured to attach to the bottom surface of the second plug body and to extend through the notch in the face plate to prevent the walkboard from lifting off the horizontal member of the scaffold when the walkboard assembly is in use.

7. The walkboard of claim 1, wherein the safety latch does not extend below a bottom plane of the walkboard.

8. A walkboard for use with a scaffold, the walkboard comprising: one or more planks forming a platform for standing, wherein the platform includes a top wall and a plurality of longitudinally extending ribs projecting downward from the top wall, and wherein the ribs define a series of longitudinally extending channels between adjacent pairs of ribs that are open at the opposing ends of the platform; an end cap assembly configured for attachment to each end of the platform, each end cap assembly comprising: one or more face plates covering the open-ended channels in the platform; and a pair of hooks extending from the one or more face plates and configured to engage a horizontal member of the scaffold, wherein the hooks are arranged on the one or more face plates such that the hooks on a first walkboard in a non-inverted orientation are offset from the hooks on a second wallboard when the second walkboard is inverted for stacking on the first walkboard.

9. The walkboard of claim 8, wherein the end cap assembly comprises a single face plate with at least two hooks.

10. The walkboard of claim 8, wherein the end cap assembly further comprises a plurality of plug bodies extending from a back of the one or more face plates and configured to fit into respective channels in the platform.

11. The walkboard of claim 10, wherein the end cap comprises: a first plug body in the plurality of plug bodies and configured to fit into a first channel in the platform, the first plug body having a first angled surface with a first mounting hole formed therein; a second plug body in the plurality of plug bodies and configured to fit into a second channel in the platform adjacent to the first channel, the second plug body having a second angled surface with a second mounting hole formed axially aligned with the first mounting hole; and an elongate fastener configured to extend through the aligned mounting hold to or secure the end cap to the platform.

- 12.** The walkboard of claim 2, wherein the elongate fastener comprises a bolt configured to extend through the first and second mounting holes and a nut configured to thread onto one end of the bolt to fasten the end cap to the first rib separating the first and second channels.
- 13.** The walkboard of claim 8, further comprising: a notch in a lower edge of the face plate; a safety latch configured to attach to the bottom surface of the second plug body and to extend through the notch in the face plate to prevent the walkboard from lifting off the horizontal member of the scaffold when the walkboard assembly is in use.
- 14.** The walkboard of claim 13, wherein the safety latch does not extend below a bottom plane of the walkboard.
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